

1. On the backend, nothing communicates until MACE models are fully downloaded, maybe we don't need to download MACE every time but just once during development.
2. Fix the UI that covers model names
3. It is not mobile responsive and things don't align on mobile
4. Aside from Open Folder, we should have let Open Folder have access to general files like in downloads
5. To write texts, it doesn't need to write a python script

Critical (6 fixes)

6. **window.undoManager exposed globally** (undo.js) — AI undo/redo tools now work
7. **Frame/energy desync fixed** (frameUtils.js, aiagent.js, main.js, fileHandler.js) — frameEnergies and lastMaceResults are now cleared everywhere xyzFrames is cleared; loadFrames() auto-syncs energies
8. **SSE JSON.parse wrapped in try/catch** (aiagent.js) — malformed chunks are skipped instead of crashing the stream
9. **Mutating AI tools serialized** (aiagent.js) — read-only tools (L1/L6) run in parallel, mutating tools (L3/L4/L5) run sequentially to prevent race conditions
10. **Feature gating fixed** (handleFeatures.js) — updateFeatureAccess() now respects the signedIn parameter instead of always enabling everything
11. **ORCA parser frame/energy mismatch** (fileHandler.js) — frameEnergies.push() now only happens when a frame is actually added

High Severity (7 fixes)

12. **Duplicate pointerdown listeners** (main.js) — old handler removed before re-adding
13. **Mismatched event listener targets** (main.js) — disableAtomInteractions/enableAtomInteractions now consistently use window for pointermove and enhancedOnPointerDown
14. **selectAtom bounds check** (main.js) — early return for out-of-range indices
15. **Missing undo snapshots** (aiagent.js) — transform_atoms and split_molecule now call saveUndoState()
16. **analyzeMolecule error handler** (index.html) — now hides the correct loading element
17. **generateAngleScan validation** (scanUtils.js) — validates atomsToMove before use
18. **Tool arguments JSON.parse** (aiagent.js) — wrapped in try/catch with graceful error return

Medium Severity (12 fixes)

19. **Hardcoded /4 → molecule.stretch** (aiagent.js) — 7 occurrences + set_bond_distance now use dynamic stretch

20. **Chart.js memory leak** (index.html) — previous chart destroyed before creating new one
21. **Thinking timer cleanup** (index.html) — setInterval cleared when stream ends without thinking_done
22. **SSE timeout timer leak** (aiagent.js) — clearTimeout called on both success and error paths
23. **streamMaceSSE buffer tail** (maceUtils.js) — remaining buffer processed after stream ends; SyntaxError properly detected
24. **Zero force treated as missing** (maceUtils.js) — if (force) → if (Array.isArray(force))
25. **Unknown file type default** (utils.js) — returns null instead of 'mol'; fileHandler shows error toast
26. **get_bonded_atoms compression** (aiagent.js) — checks result.bonds instead of result.data
27. **postJson body override** (apiUtils.js) — body is now set last so options can't overwrite it
28. **Input validation** (aiagent.js) — 6 tools now validate params.indices, params.element, params.atomsToMove
29. **Three.js memory leaks** (main.js) — env texture disposal on switch; recreateRenderer calls dispose(); uses enhancedOnPointerDown after recreate
30. **parsedData null check** (fileHandler.js) — graceful error if parsing produces no atoms; frame slider max clamped to 0 minimum
- 31.